



BSc (Hons) in **Information Technology**
Specialized in AI

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 02

Objective & Lecture Mapping: In Lecture 03, we learned that a mathematically perfect "Table-Driven Agent" is impossible to build due to combinatorial explosion. Instead, we must write Agent Programs. Today, you will implement a **Simple Reflex Agent** using Condition-Action rules, observe its fatal flaws in a partially observable environment, and then upgrade it to a **Model-Based Agent** that maintains an internal memory state.

Part 1: Practical Implementation — Coding the Architectures

Step 1.1: Setting the Trap (Partial Observability)

To see why Simple Reflex agents fail, we must handicap your agent's sensors. We are moving from a Fully Observable world to a Partially Observable one.

1. Open your `visual_grid_game.py` from Lab 01.
2. Locate the `get_percept(self)` method.
3. Modify it so it **no longer returns** the exact global coordinates (`agent_pos`). Instead, it should only return local booleans, e.g., `{ 'wall_ahead': True/False, 'food_here': True/False }` based on checking the adjacent cell in its current facing direction.

Step 1.2: The Simple Reflex Agent (Implementation & Failure)

1. Create a new class called `SimpleReflexAgent` .
2. Implement a `sense_and_act(self, percept)` method that uses strictly IF-THEN logic (Condition-Action rules). *Do not use an `__init__` method to store history.*

Example Logic: IF food_here THEN suck; IF wall_ahead THEN turn_left; ELSE move_forward

3. Run the simulation. **Observe:** Watch the agent get trapped in a corner or a U-shaped wall, infinitely repeating a cycle (e.g., turn left, step forward, hit wall, turn left...).

Step 1.3: The Model-Based Agent (Memory & State)

1. Create a new class called `ModelBasedAgent`.
2. Implement an `__init__(self)` method to initialize an internal memory state (e.g., `self.visited_cells = set()` or a relative movement tracker).
3. In `sense_and_act(self, percept)`, first **update the state** (Transition & Sensor Model) by recording the current percept and the last action taken.
4. Update your IF-THEN rules to query the memory.
Example Logic: IF wall_ahead AND left_is_visited THEN turn_right
5. Run the simulation. **Observe:** The agent should now remember where it has been, realize it is in a loop, and choose an alternate path to escape.

Part 2: Theoretical Evaluation & Lecture Mapping

Based on your practical observations above, answer the following questions to map your code directly to the architectural theories covered in Lecture 02.

1. (Remember) According to Lecture 02, why is it impossible to program a mathematically perfect "Table-Driven Agent" for complex environments like Chess? What happens as the agent's lifetime increases?

A Table-Driven Agent is impossible for complex environments like Chess because it would require storing every possible state and action combination. As the agent's lifetime increases, the number of possible situations grows exponentially, making the table too large to store and manage.

2. (Understand) Look at the code you wrote for your `SimpleReflexAgent` . Identify and explain the specific lines of code that represent the "Condition-Action Rules" discussed in the lecture.

- if percept["food_here"]: → Condition: If food is detected → Action: Suck.
- elif percept["wall_ahead"]: → Condition: If a wall is ahead → Action: Turn left.
- else: → Condition: No food or wall detected → Action: Move forward.

These rules show that the Simple Reflex Agent selects actions only based on the current percept without using memory.

3. (Analyze) Your `SimpleReflexAgent` likely got stuck in an infinite loop during Step 1.2. Based on the lecture, analyze exactly why this happened. How did the combination of "Partial Observability" and a lack of "Percept History" cause this failure?

The `SimpleReflexAgent` got stuck because it only uses the current percept and has no percept history or memory. In a partially observable environment, the agent cannot see the complete state of the world. When it reaches a situation it has already encountered, it cannot remember that it was there before.

Without percept history:

The agent repeats the same condition-action rule every time.

It cannot recognize that it is moving in a loop.

It keeps performing the same actions (e.g., turn left, move forward, hit wall).

Therefore, the combination of limited information (partial observability) and no memory of previous states (lack of percept history) causes the agent to fail and become trapped in infinite cycles.

4. (Evaluate) In Step 1.3, you added an internal state to your `ModelBasedAgent` . Evaluate how your specific code handles the "Transition Model" (how the world evolves) and the "Sensor Model" (how the agent's actions affect the world).

In my ModelBasedAgent, the Sensor Model is handled by the `get_percept()` method. It converts the actual grid environment into percepts such as `wall_ahead`, `food_here`, and `trap_here`, allowing the agent to understand its surroundings.

The Transition Model is handled by the `execute_action()` method. It updates the environment after actions by moving the agent, changing the score, collecting food, and applying trap penalties.

Strengths: Both models are deterministic and efficient. The agent receives accurate sensor information, and actions produce predictable changes in the environment.

Weaknesses: The models do not handle uncertainty. They cannot adapt to noisy sensors or unexpected actions, such as incorrect perceptions or random movement effects.

Finalize and Lock Lab 02 Documentation

Incomplete: 0/11 questions answered. Please provide detailed responses for all questions.



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